

Literature Status: Singularity Probability of Random Sign Matrices

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Status: literature already answered. Rudelson and Vershynin record the conjectured optimal exponential rate $(1/2 + o(1))^n$ for singularity of an i.i.d. random sign matrix. Tikhomirov proves this exact asymptotic in arXiv:1812.09016, published in the *Annals of Mathematics* in 2020.

1 Original conjecture

M. Rudelson and R. Vershynin, *The Littlewood–Offord Problem and invertibility of random matrices*, arXiv:math/0703503 [1], discuss random $n \times n$ matrices with independent symmetric $\{-1, 1\}$ entries in Section 1.1. On arXiv PDF page 3 they call estimation of the singularity probability a longstanding open problem and record the conjectured optimal base $1/2 + o(1)$.

Equivalently, if M_n has independent random-sign entries, the conjecture is

Source conjecture.

$$\mathbb{P}\{M_n \text{ is singular}\} = \left(\frac{1}{2} + o_n(1)\right)^n.$$

Their paper improves and develops least-singular-value estimates, but it does not prove this sharp singularity asymptotic.

2 Separate later answer

K. Tikhomirov, *Singularity of random Bernoulli matrices*, arXiv:1812.09016, published in *Annals of Mathematics* **191** (2020), 593–634 [2], states in its abstract and equation (1) on arXiv PDF page 1:

Later theorem. For an $n \times n$ matrix M_n with independent $\{-1, 1\}$ entries,

$$\mathbb{P}\{M_n \text{ is singular}\} = \left(\frac{1}{2} + o_n(1)\right)^n.$$

The paper explicitly says that this confirms the conjecture. Its Theorem A is formulated more generally for Bernoulli(p) zero–one matrices and a rank-one shift. Taking $p = 1/2$ and shift $s = -1/2$ produces a matrix with independent entries $\pm 1/2$; multiplying by 2 preserves singularity and gives the random-sign model above.

Thus the precise open conjecture recorded in arXiv:math/0703503 is completely answered by a separate later paper. No new theorem is claimed here.

3 Search scope

A bounded check through 9 August 2026 used the run registry, exact source wording, the paper title, and searches for the sharp Bernoulli singularity rate. The decisive match is arXiv:1812.09016 and its published Annals version; both state the exact asymptotic explicitly.

References

- [1] M. Rudelson and R. Vershynin, *The Littlewood–Offord Problem and invertibility of random matrices*, Adv. Math. **218** (2008), 600–633; arXiv:math/0703503.
- [2] K. Tikhomirov, *Singularity of random Bernoulli matrices*, Ann. of Math. (2) **191** (2020), 593–634; arXiv:1812.09016. DOI: 10.4007/annals.2020.191.2.6.